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SECR1013 DIGITAL LOGIC

MODULE 8: COUNTERS part A: Intro & Async Counters

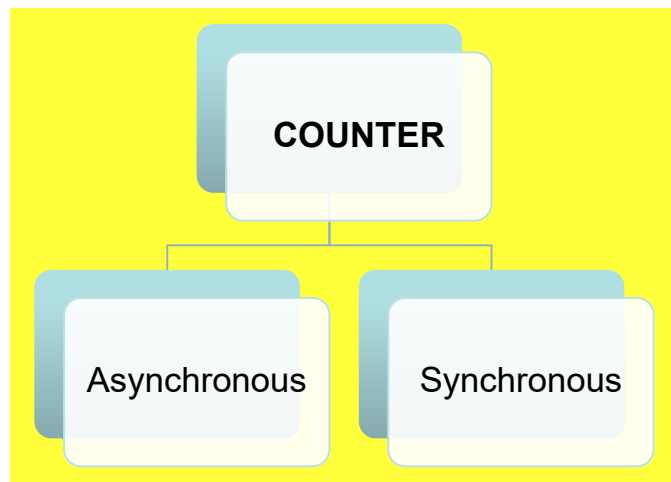
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The objectives of this module are:

1. To distinguish the different between synchronous and asynchronous counters.
2. To illustrate the designing of counters .
3. To explain the application of counters.
4. To demonstrate the analysis of sequential circuits.

- There is a variety of counters based on its construction.
- **Counter** is type of *sequential logic circuit*.
- In general, two categories of counters:



Both categories can be differentiated by criterion:

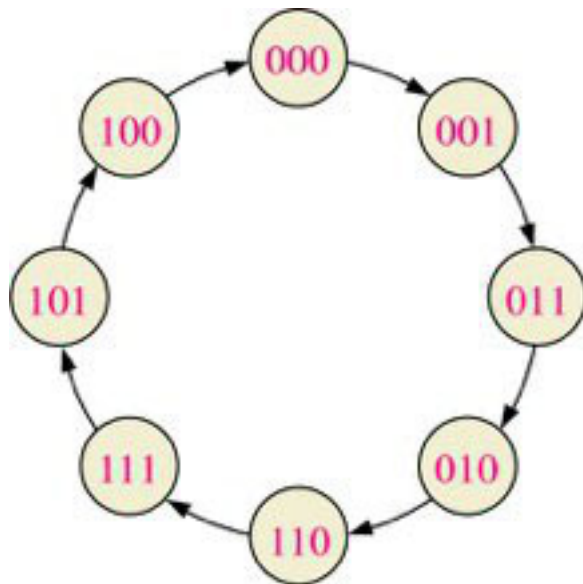
1. Clock Trigger: Positive edged or Negative edged
2. Counts: Binary, Decade
3. Count Direction: Up, Down, or Up/Down

In the previous lectures, the characteristic of S-R, D, J-K and T flip-flops were discussed. These flip-flops can be connected together to perform certain operations.

In the following lectures, we will use these flip-flops to construct a variety of counters. In addition we will also learn the method for analysing the different types of counters.

Introduction: Counters

- A **counter** → Any sequential circuit that goes through a prescribed **sequence of states** upon the application of **input pulses**.
- The **sequence of states** in a counter may follow a binary count or any other sequence.

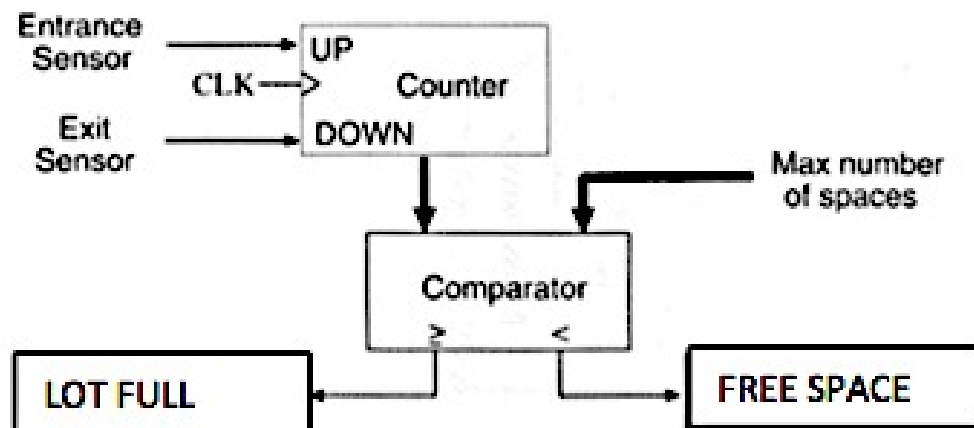


- The states changed by the **input pulse**;
- the input pulses (count pulses) may be clock pulses or they may originate from an external source; may occur at prescribed intervals of time or at random.

continue...

Uses of Counters

- To count the number of times that a certain event takes place; the occurrence of event to be counted is represented by the input signal to the counter (see figure below).





- To control a fixed sequence of actions in a digital system
 - e.g. gate A must be open before gate B.



- To control a fixed sequence of actions in a digital system
 - e.g. gate A must be open before gate B.
- To generate timing signals
 - e.g. a bomb can only be detonated 20 second after a button is pressed



- To generate clocks of different frequencies.
 - By dividing the frequency, we can get a different clock frequencies from the same source

There are many devices used that can tell the time without having to be manually programmed



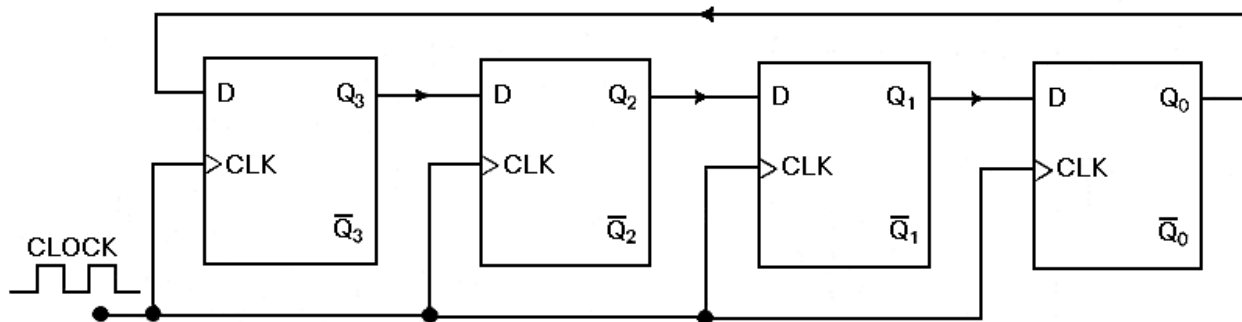
Characteristic of Counter

- Counter can be described by the following characteristics:
 1. Asynchronous or synchronous
 2. Counting sequence
 3. Modulus (MOD)
 4. Whether the counter repeat the counting sequence

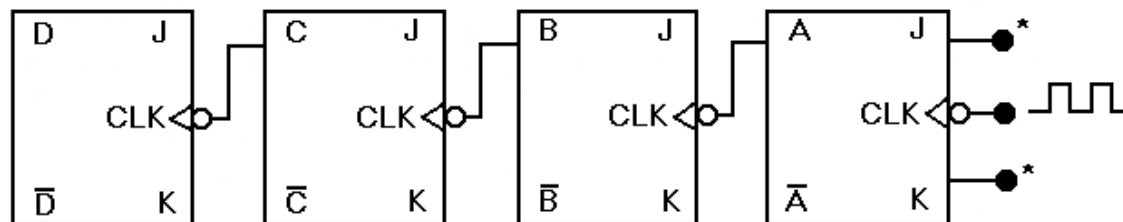
Counters: Characteristics

1. Asynchronous / Synchronous

- Whether the clock come from the same source or not
 - Synchronous – all clock from the same source
 - Asynchronous – clock for each flip-flop comes from a different source



Synchronous Counter
(clock from same src)



Asynchronous Counter
(clock from diff src)

* All J and K inputs assumed to be at logic 1.

2. Counting Sequence: Up | Down | Up-Down

- Up or Down – unidirectional – the count sequence only one way
- Up-Down – bidirectional – the direction of the count can be change during operation
- Because of limited word length, the count sequence is limited.
 - For an n -bit counter, the range of the count is $[0, 2^n-1]$.
 - The count sequence usually repeats itself.

- When **counting up**, the count sequence goes in this manner:
 $0, 1, 2, \dots 2^n-2, 2^n-1, 0, 1, \dots$ etc.
- When **counting down** the count sequence goes in the same manner:
 $2^n-1, 2^n-2, \dots 2, 1, 0, 2^n-1, 2^n-2, \dots$ etc.

3-bit Up Counter	3-bit Down Counter
000	000
001	111
010	110
011	101
100	100
101	011
110	010
111	001

3. Modulus (MOD):

- Modulus (MOD): the number of state that the counter can have.
 - e.g. MOD 4 (4 states with the following state 0,1,2,3), MOD 9 (0,1,2,...,8).
 - the maximum count usually $2^N - 1$, where N is the number of the FF, or for the truncated sequence the max count will be $< 2^N - 1$.

Example:

Maximum count = $16 = 2^4$

Normal count $\rightarrow 0, 1, 2, \dots (2^4-1) = 15$

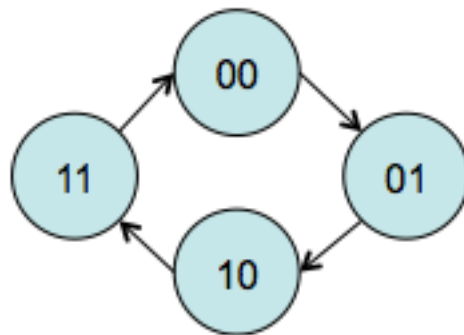
Truncated **MOD 9** $\rightarrow 0, 1, 2, \dots 8$

maximum no of
state

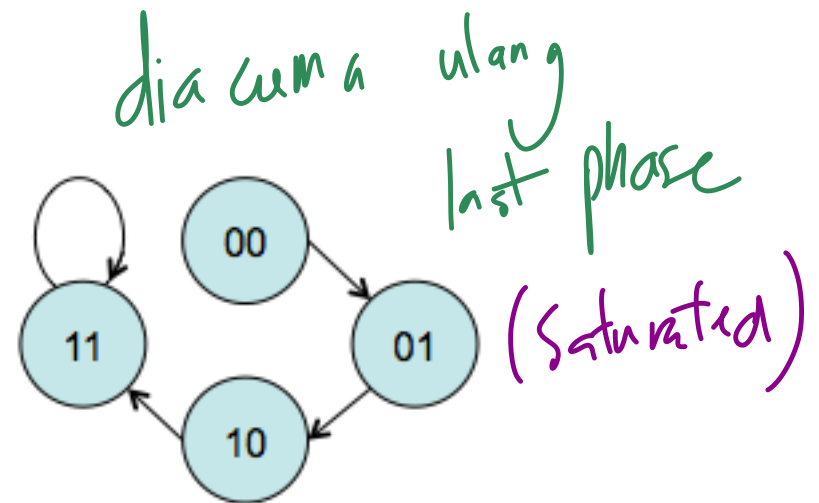
$$2^n - 1$$

4. Repeat of Counting Sequence:

- Whether the counter repeat the counting sequence:
 - Recycle – if the counter goes back to the first count after the maximum count
 - Saturated – the counter repeat the maximum count if count up or repeat the minimum if count down



Recycle



Saturated



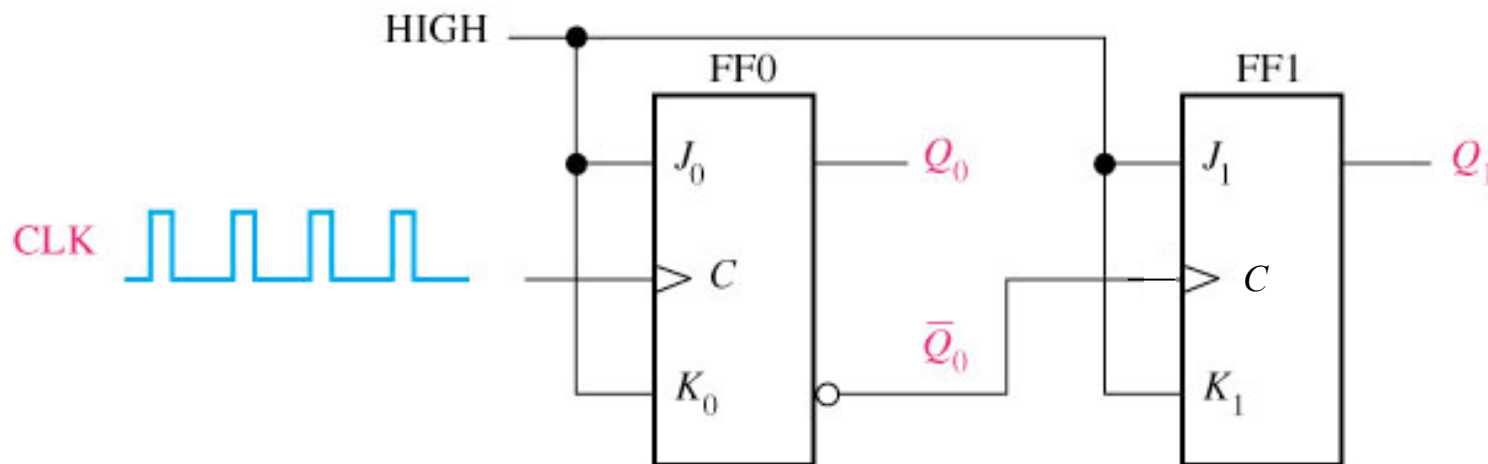
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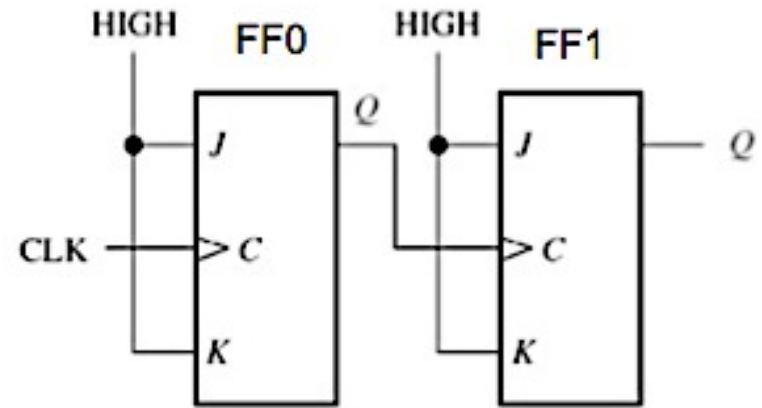
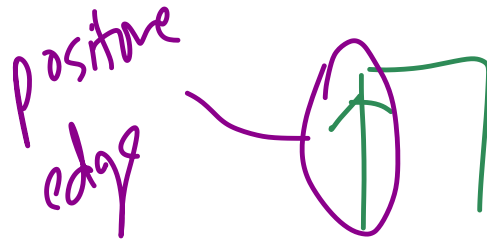
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Asynchronous Counters

Asynchronous Counter (AC)

- An AC is one in which the flip-flop (ff) within the counter do not change states at exactly the same time because they **do not have a common clock pulse**.
- The **clock input** of an AC is always connected only to the **LSB ff**
- An AC also known as **ripple counter** – message (signal) passing or propagation delay.





Asynchronous counter operation:

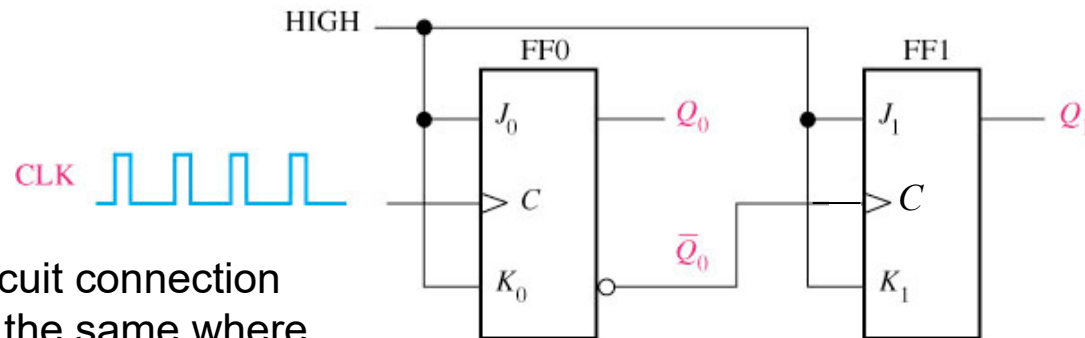
Asynchronous

- The external clock is connected to the clock input of the first flip-flop (FF0) only.
- So, FF0 changes state at **positive edge** of each clock pulse, but FF1 changes only when triggered by the positive edge of the Q output of FF0.
- Because of the inherent propagation delay through a flip-flop, the transition of the input clock pulse and a transition of the Q output of FF0 can never occur at exactly the same time.
- Therefore, the flip-flops cannot be triggered simultaneously, producing an **Asynchronous operation**

This asynchronous counter is slow because the cascaded clocking scheme

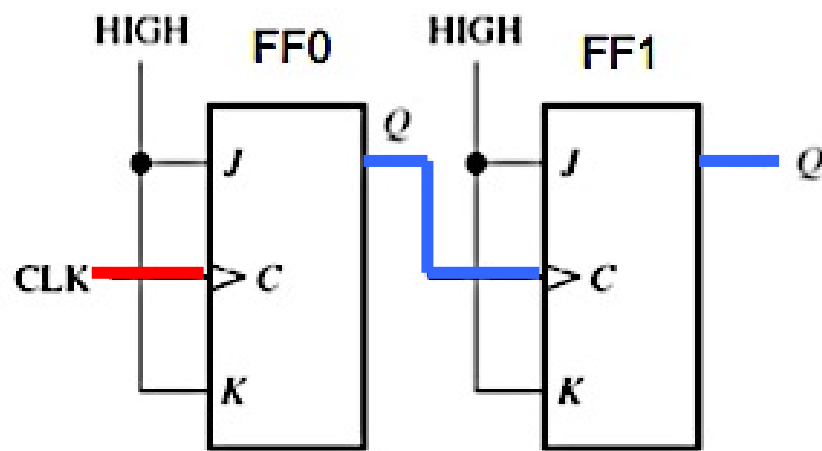
- The clock source ripples from stage-to-stage
- The ripple effect is similar to that of a ripple carry adder circuit

Asynchronous Counter operation:

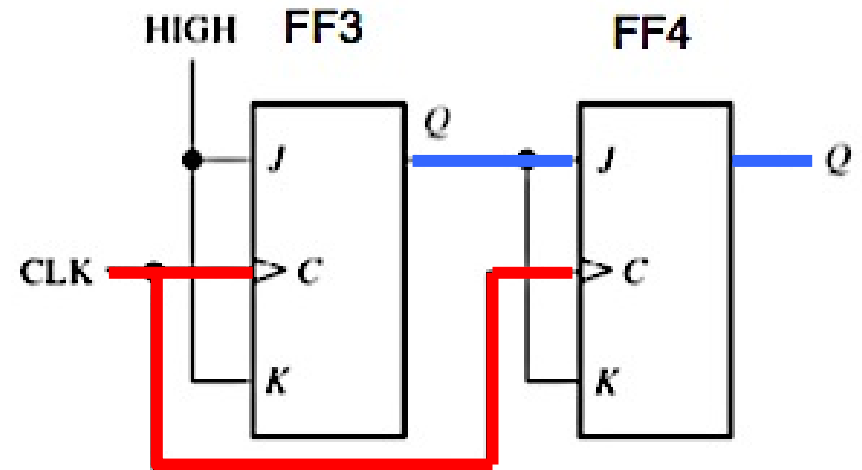


1. The circuit connection almost the same where external clock connected to **LSB** Flip Flop (FF0)
2. Clock for each FF come from the previous FF, **except** for the first FF (i.e., FF0)
3. Every FF operates in **Toggle** mode; i.e., next output is a complement of the previous output.
4. The design **connections is the same** for number of bits of input; more bits → add more FFs
5. The difference in connection will determine on:
 - to count UP or DOWN
 - type of FF – positive edge OR negative edge triggered

- Comparison of asynchronous & synchronous.
- See the different clock connection !



Asynchronous



Synchronous

Generalization of an Asynchronous Counter

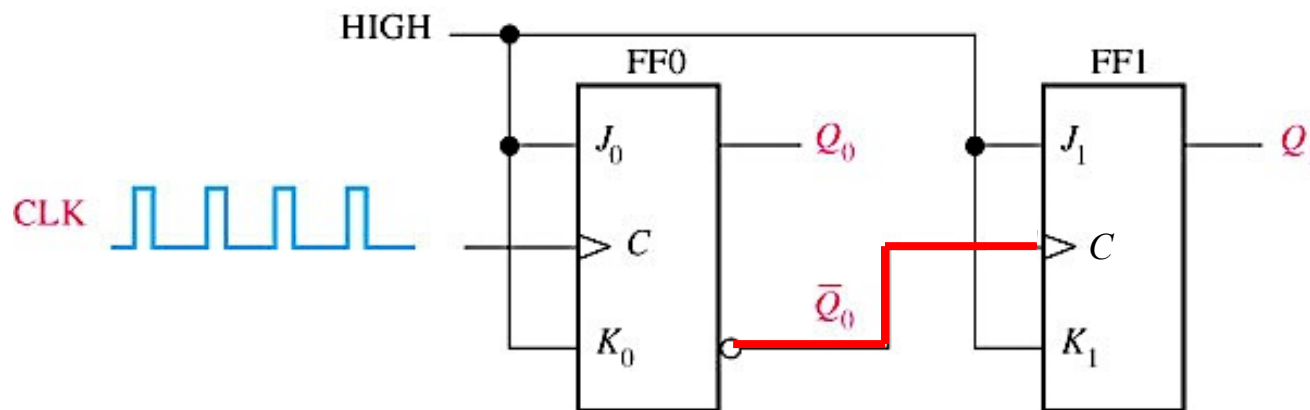


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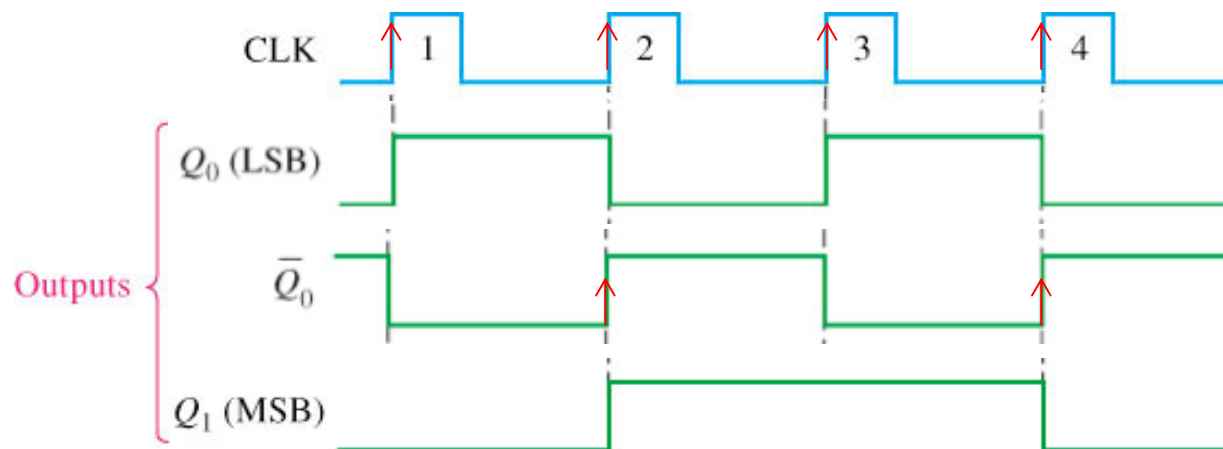
	Positive Edge	Negative Edge
Count Up		
Count Down		



Count Up 2-bit AC



- Positive edge
- Assume that Q_0 initially LOW



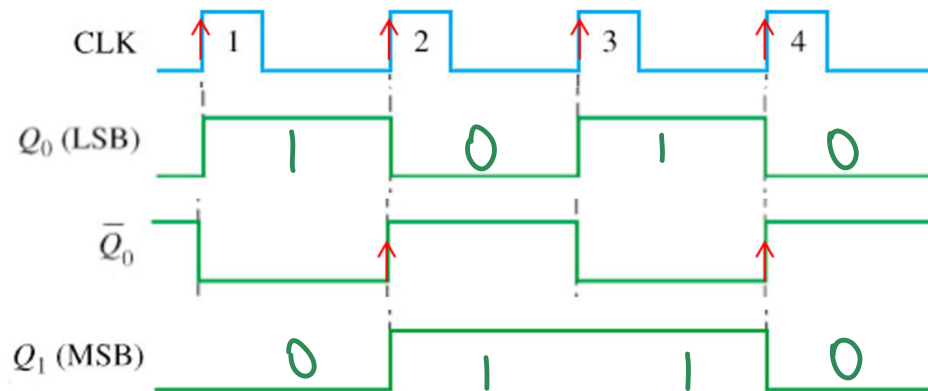
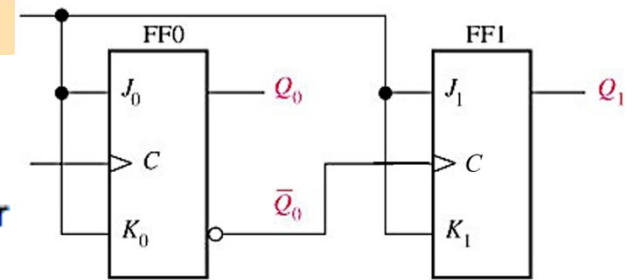
inverse of 2.

- Output of Q_0 is based on CLK
- Output of Q_1 is based on $\overline{Q_0}$

depends on $\overline{Q_0}$

Notice:

1. Input JK for each FF is tied up together. Therefore $J=K=1$. In other words, **Toggle mode** i.e. every clock pulse Q will be complemented.
2. The clock of FF1 comes from the complemented output of FF0, $\bar{Q}_0$.
3. For FF0, every positive edge of the clock the output, Q_0 will toggle.
4. For FF1, the clock depends on $\bar{Q}_0$ and the output Q_1 will only toggle on the positive edge of $\bar{Q}_0$.
5. The output of the circuit counter is read $Q_1 Q_0$.



	MSB	LSB
CLOCK PULSE	Q_1 (MSB) (Bit 2)	Q_0 (LSB) (Bit 1)
Initially	0	0
1	0	1
2	1	0
3	1	1
4	0	0

The 2-bit ripple counter circuit above has four different states, each one corresponding to a count value. Similarly, a counter with n flip-flops can have 2^n states. The number of states in a counter is known as its

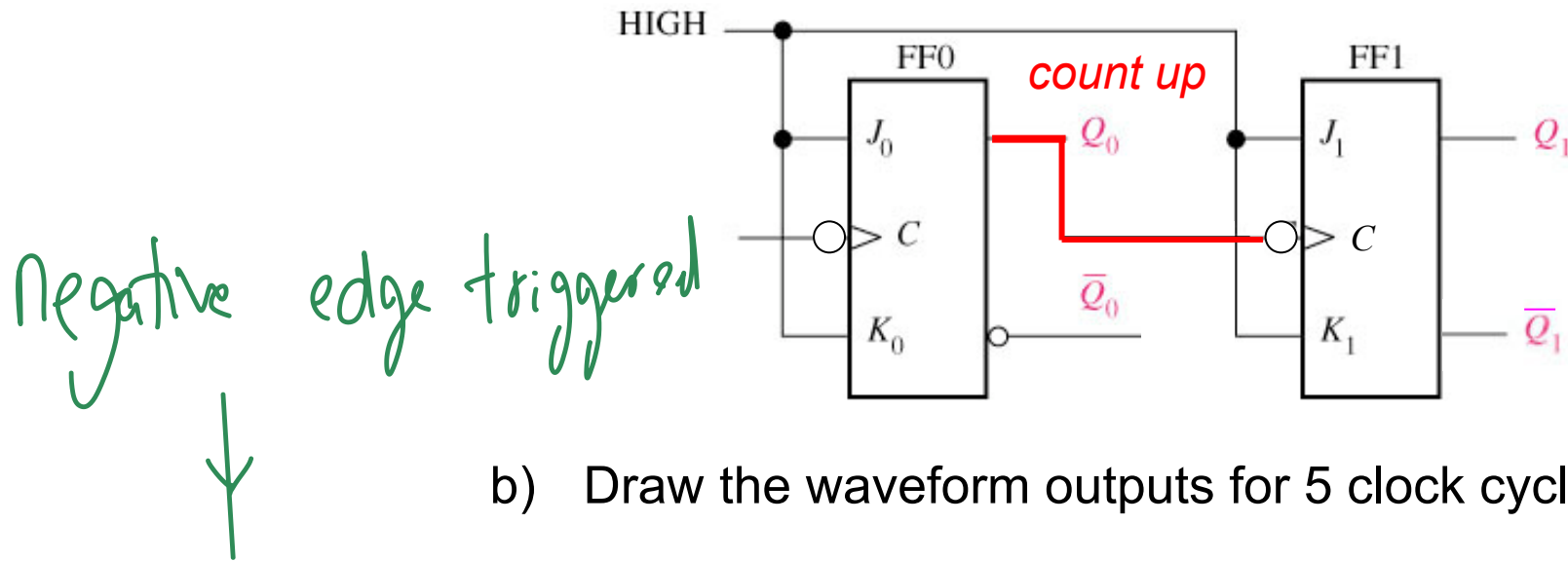
MOD (modulo). Thus a 2-bit counter is a mod-4 counter.

Exercise 8a.1:

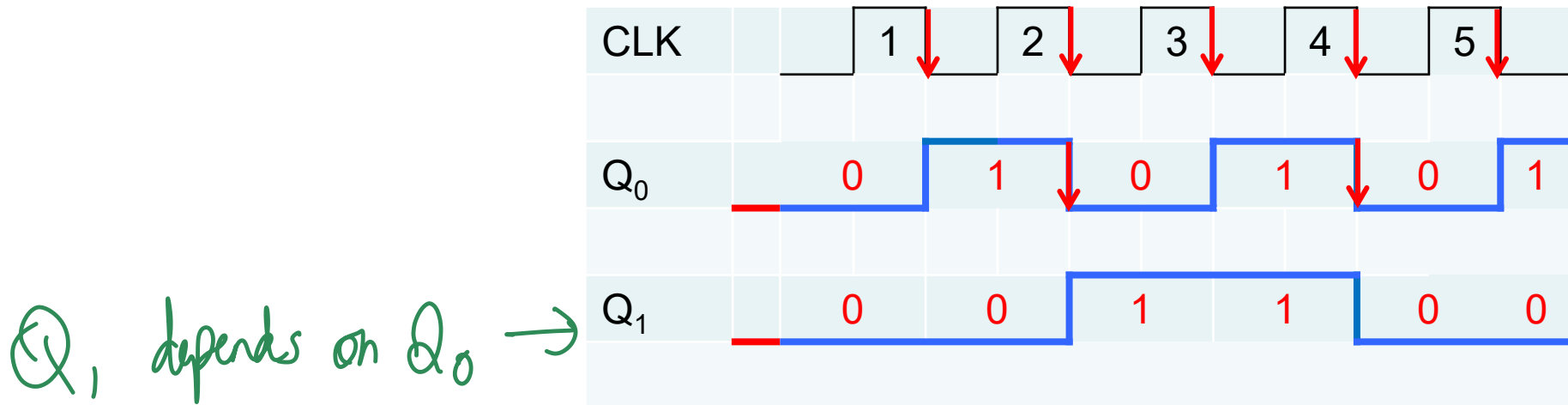
A 2-bit count up ripple counter is designed using J-K flip-flop with negative edge triggered clock.

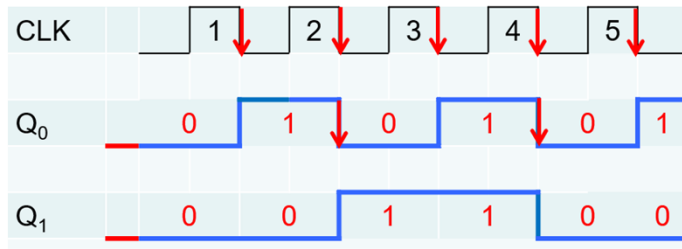
- a) Draw the connection of logic symbol.
- b) Draw the waveform outputs for 5 clock cycles.
- c) Construct a state table for the counter for 5 clock cycles.
- d) Draw the state diagram the the counter.

Solution 8a.1: a) Draw the connection of logic symbol.



b) Draw the waveform outputs for 5 clock cycles.

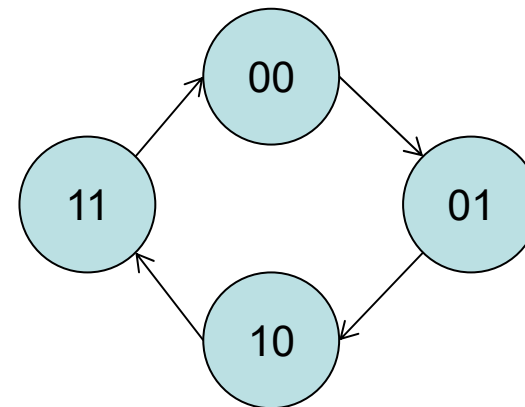




c) Construct a state table for the counter.

Clock Pulse	Q_1	Q_0
Initial	0	0
1	0	1
2	1	0
3	1	1
4	0	0
5	0	1

d) Draw the state diagram the the counter.



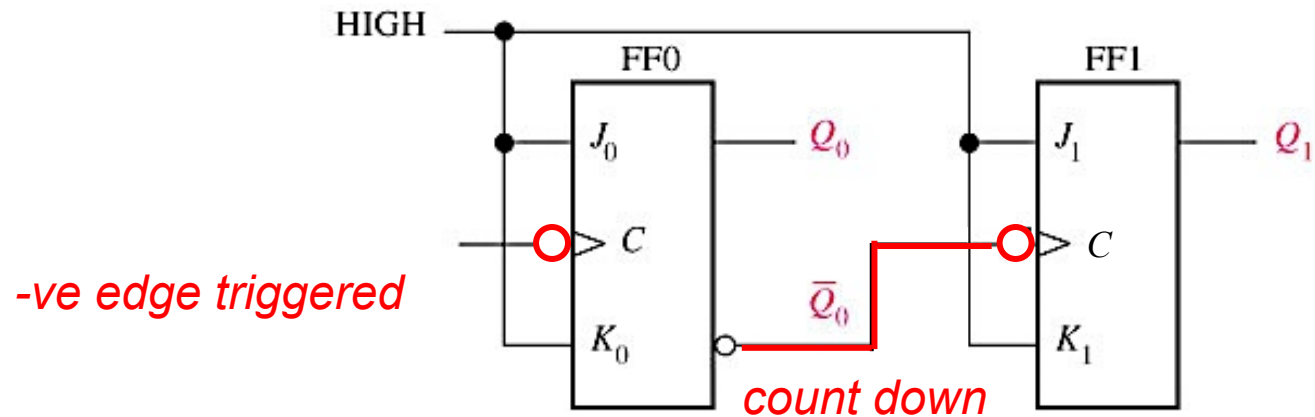
Exercise 8a.2:

A 2-bit **count down** ripple counter is designed using J-K flip-flop with **negative edge triggered** clock.

↳ add bubble @ CLK

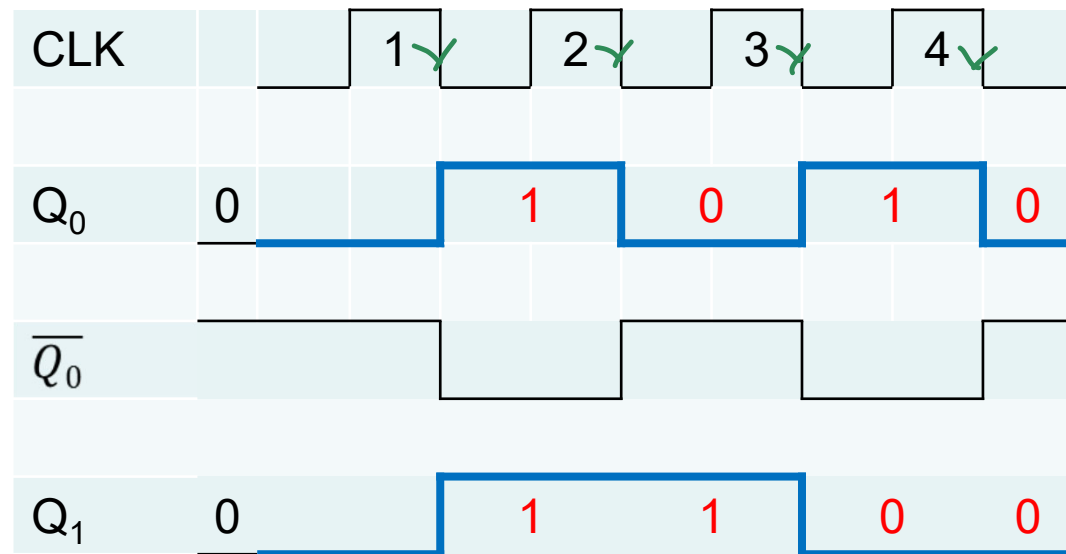
- Draw the connection of logic symbol.
- Draw the waveform outputs for 4 clock cycles.
- Construct a state table for the counter.
- Draw the state diagram the the counter.

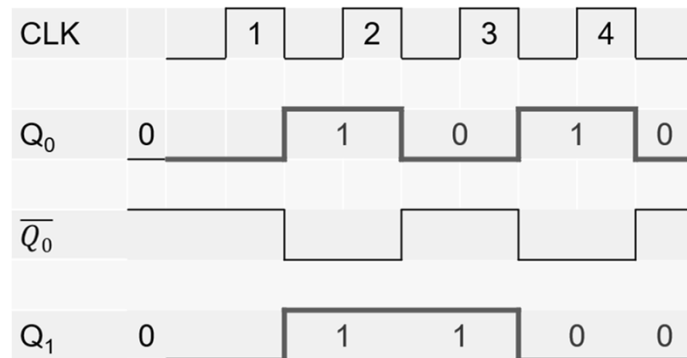
Solution 8a.2: a) Draw the connection of logic symbol.



b) Draw the waveform outputs for 4 clock cycles.

negative edge trig $\rightarrow$
 Inverse of Q_0 $\rightarrow$
 depends on $\overline{Q_0}$ $\rightarrow$

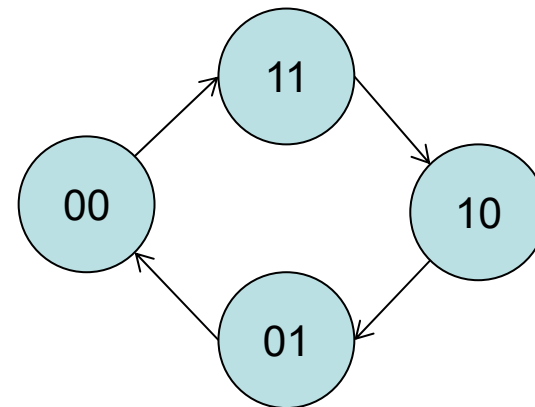




c) Construct a state table for the counter.

Clock Pulse	Q_1	Q_0
Initial	0	0
1	1	1
2	1	0
3	0	1
4	0	0

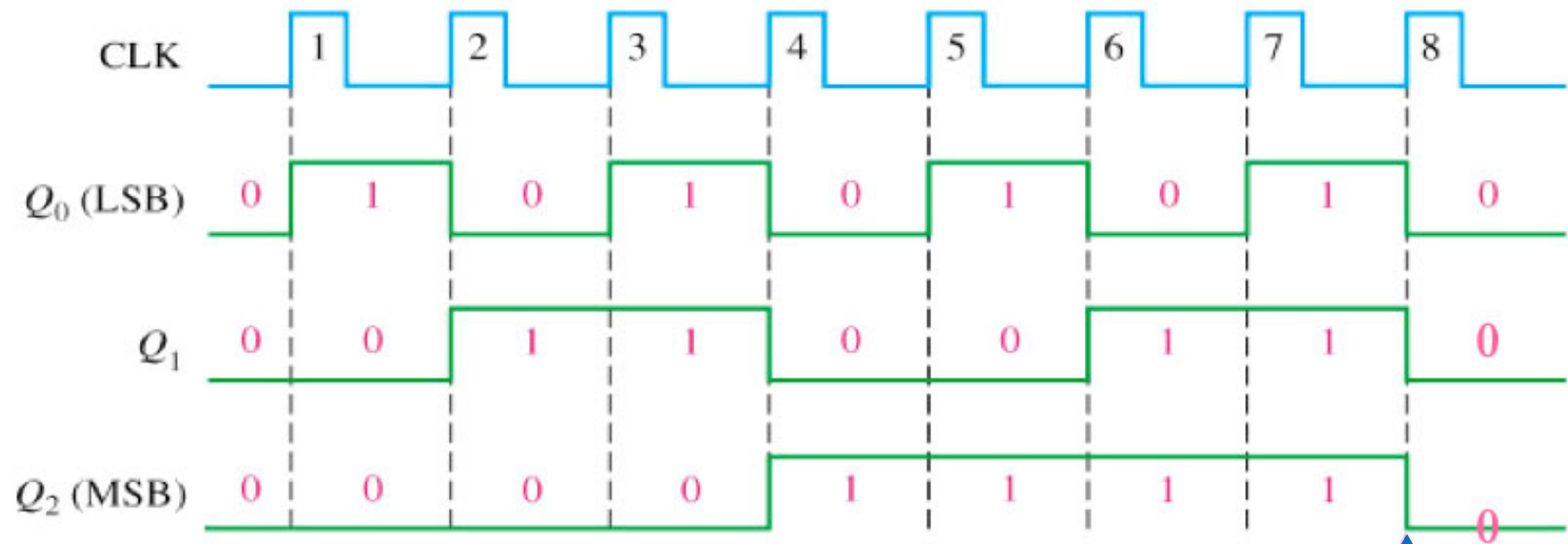
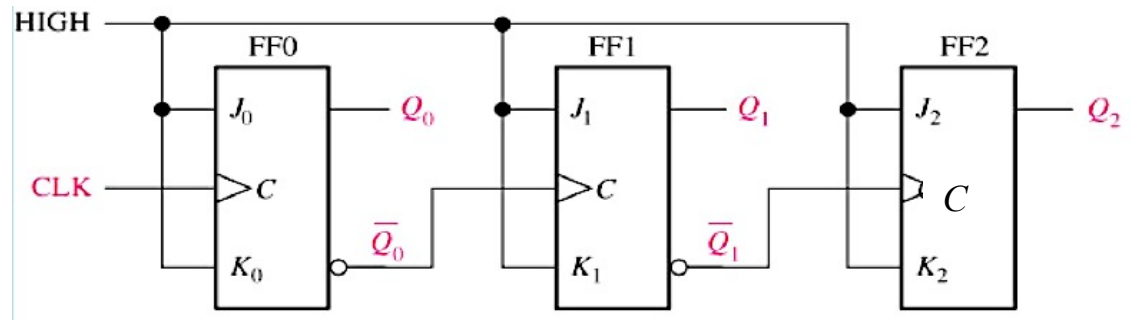
d) Draw the state diagram the counter.



Count Up 3-bit AC

- The circuit connection is the same as 2 bit asynchronous counter, for a 3 bit counter just add another flip-flop.

- Positive edge
- Assume that Q_0 initially LOW



Recycle back to 0

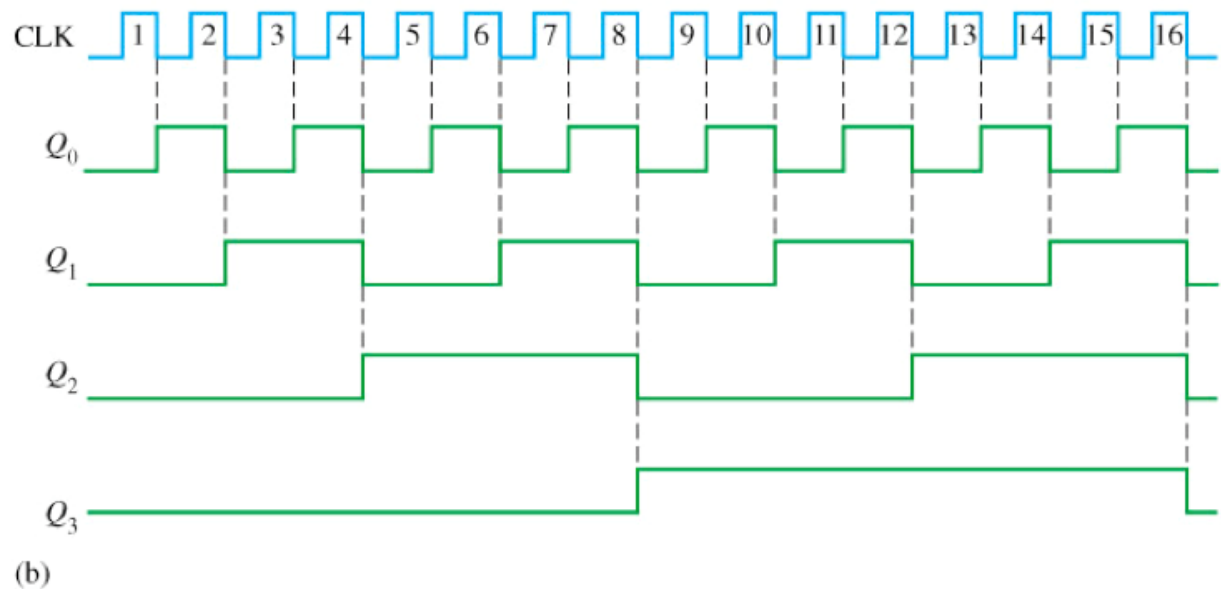
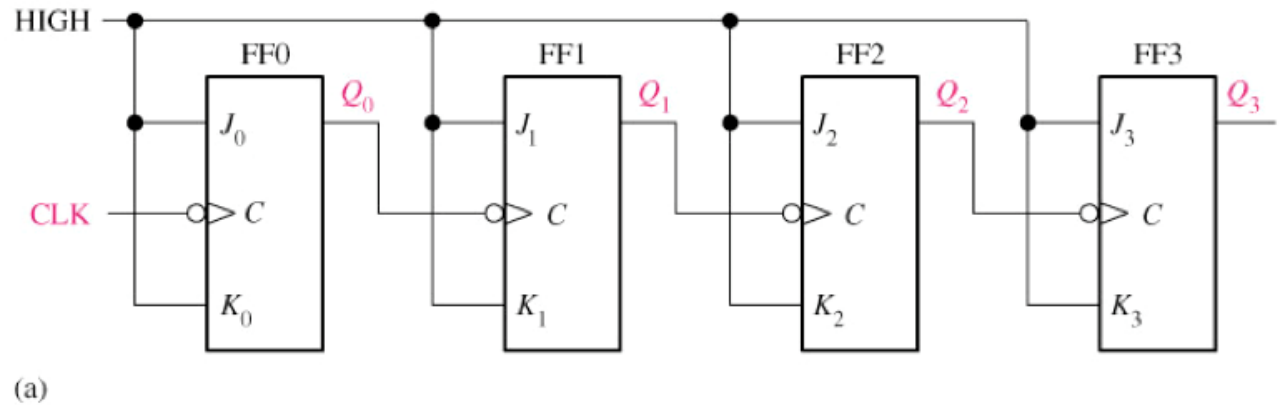
The timing diagram of the 3 bit asynchronous counter.

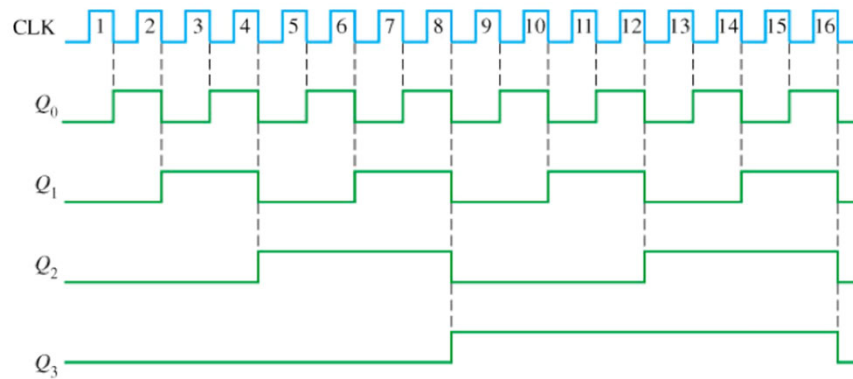
- The count change every positive edge of a clock.
- The counting sequence is 0,1,2,3,4,5,6,7 and recycle back to 0

CLOCK PULSE	Q_2 (MSB) (Bit 3)	Q_1 (Bit 2)	Q_0 (LSB) (Bit 1)
Initially	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0

Count Up 4-bit AC

- The counter using a negative edge JK FF.
- Count up counter because clock is connected to Q.





CLOCK PULSE	Q ₃ (MSB) (Bit 4)	Q ₂ (Bit 3)	Q ₁ (Bit 2)	Q ₀ (LSB) (Bit 1)
Initially	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1
16	0	0	0	0

Exercise 8a.3:

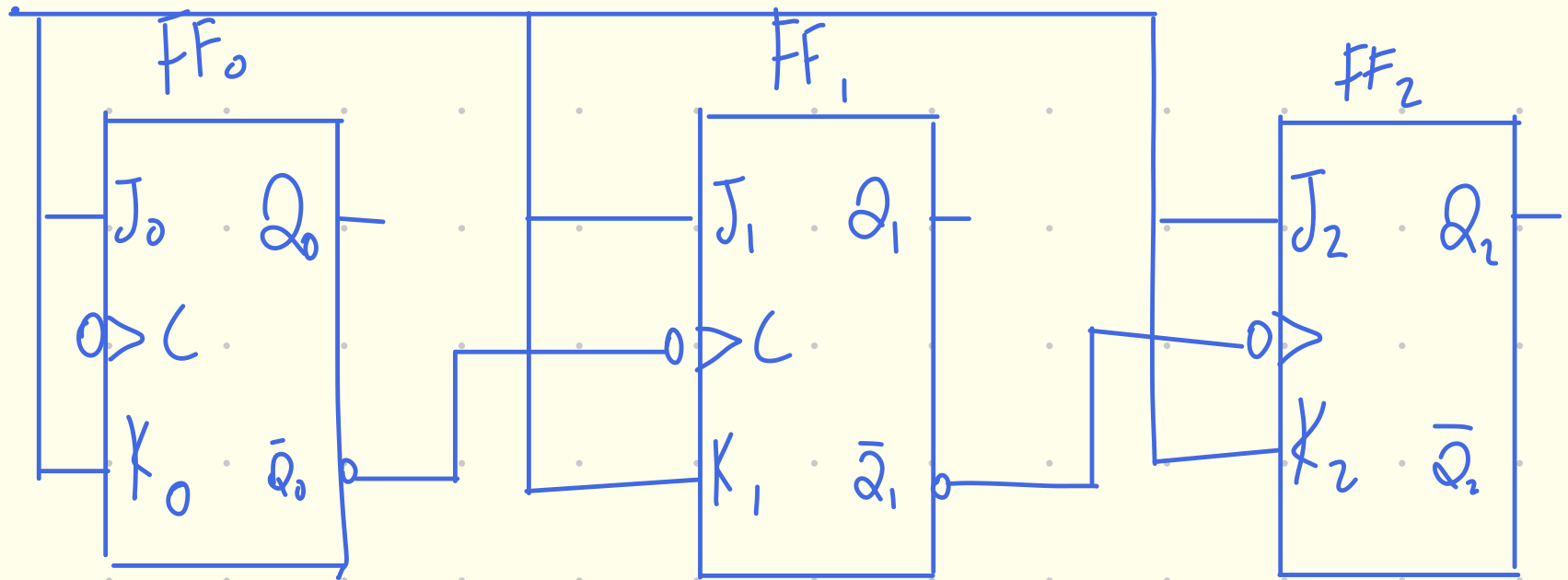


A 3-bit **count down** ripple counter is designed using J-K flip-flop with **negative edge** triggered clock.

- a) Draw the connection of logic symbol.
- b) Draw the waveform outputs for 8 clock cycles.
- c) Construct a state table for the counter.
- d) Draw the state diagram the the counter.

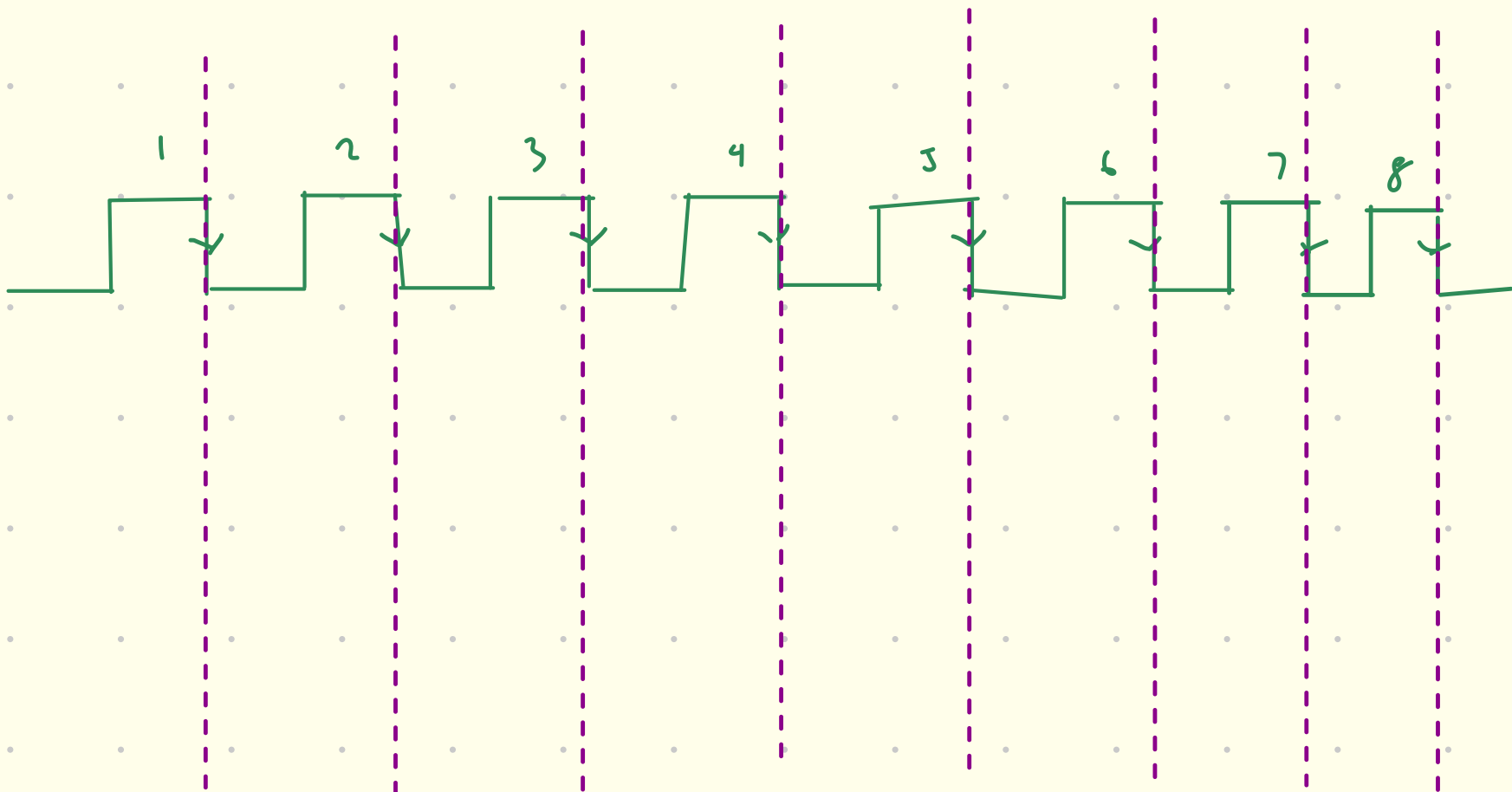
Exercise 8a.3

a) High



b)

CLK

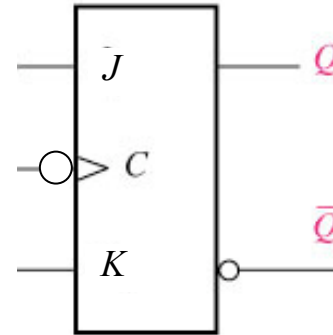


Exercise 8a.4:

A 4-bit count down ripple counter is designed using J-K flip-flop with positive edge triggered clock.

- a) Draw the connection of logic symbol.
- b) Draw the waveform outputs for 8 clock cycles.
- c) Construct a state table for the counter.
- d) Draw the state diagram the the counter.

Exercise 8a.4b:



Home
work

A 3-bit count up asynchronous counter is designed using the given above logic symbol of a flip-flop.

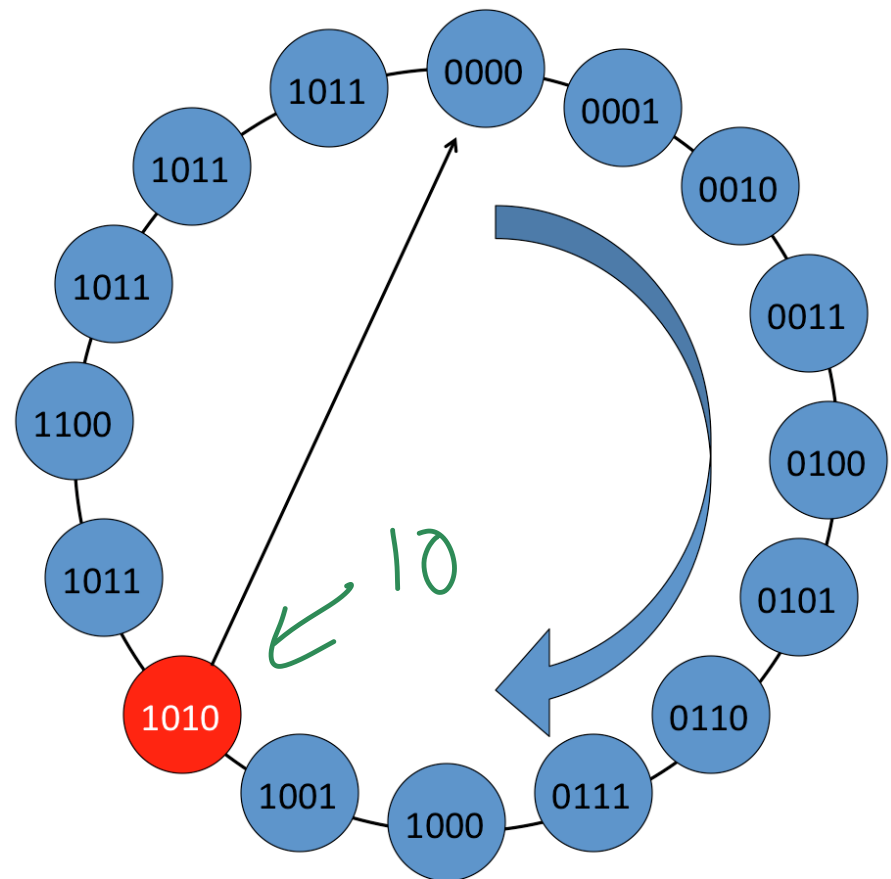
- Draw the design of the connection.
- Draw the waveform outputs for 8 clock cycles.
- Construct a state table for the counter.
- Draw the state diagram the the counter.

Asynchronous Decade Counter

- Maximum number of possible state (max. modulus) of a counter is 2^n .
- However, it can be designed less than $2^n \rightarrow$

Truncated Sequence

- One common modulus for counters with truncated sequences is ten $\rightarrow$ **decade counter**
- It will force the counter to cycle before going through all its normal states.

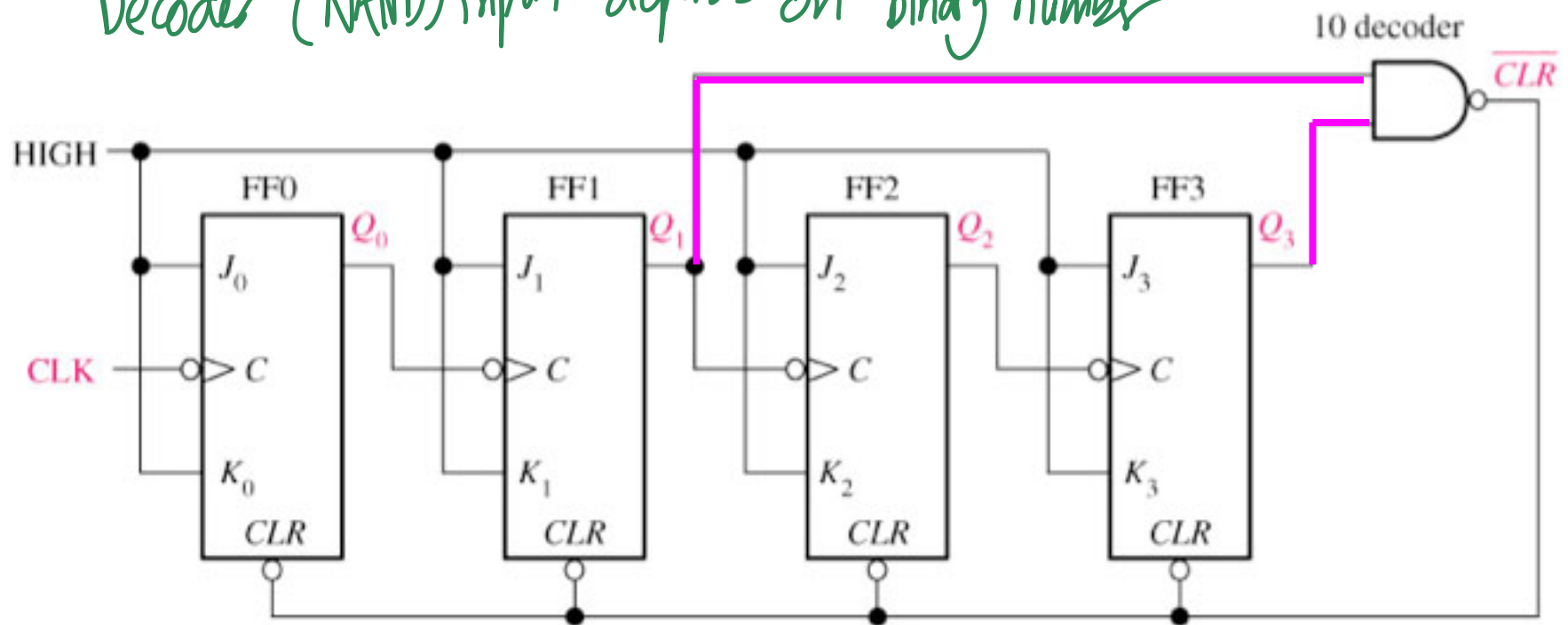
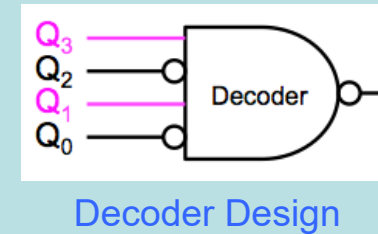


Asynchronously Clocked Modulus-10

Decode ten (1010_2) with a NAND gate and connect the output to the CLEAR inputs of all flip-flops.

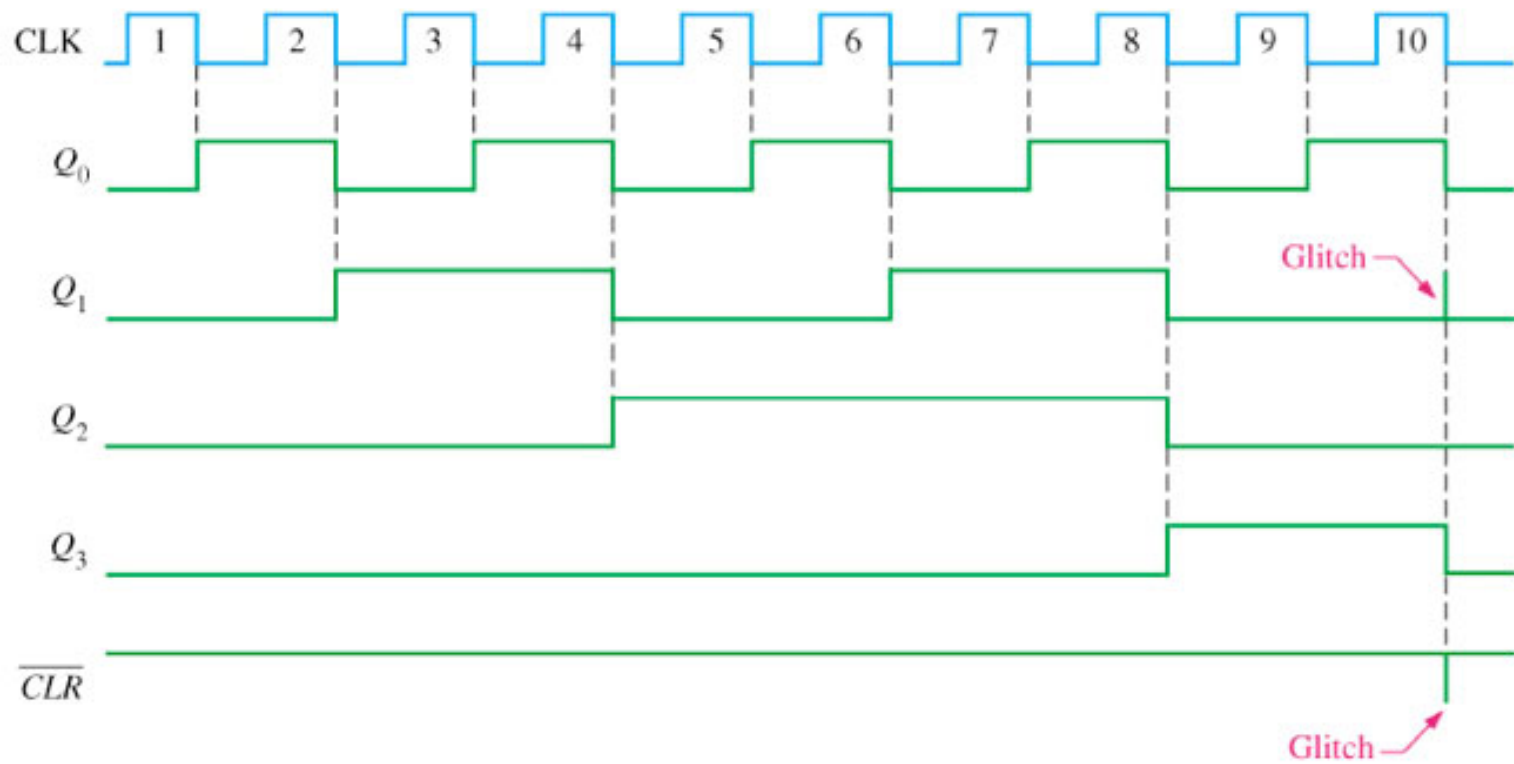
Decoder (NAND) input depends on binary number

MSB
 Q_3 Q_2 Q_1 Q_0 LsB
1 0 1 0

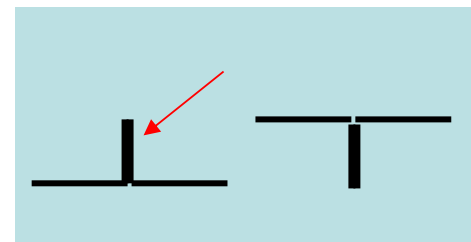


continue...

Figure: Count Up 4-bit Asynchronous Decade Counter



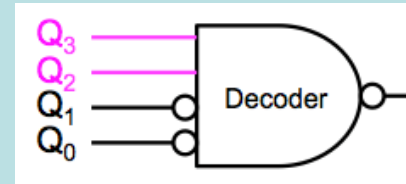
Notice that there is a **glitch** in Q_1 . The reason of this glitch is that Q_1 must first go HIGH before the count 10 can be decoded. Several nanoseconds after the decoding gate goes LOW.



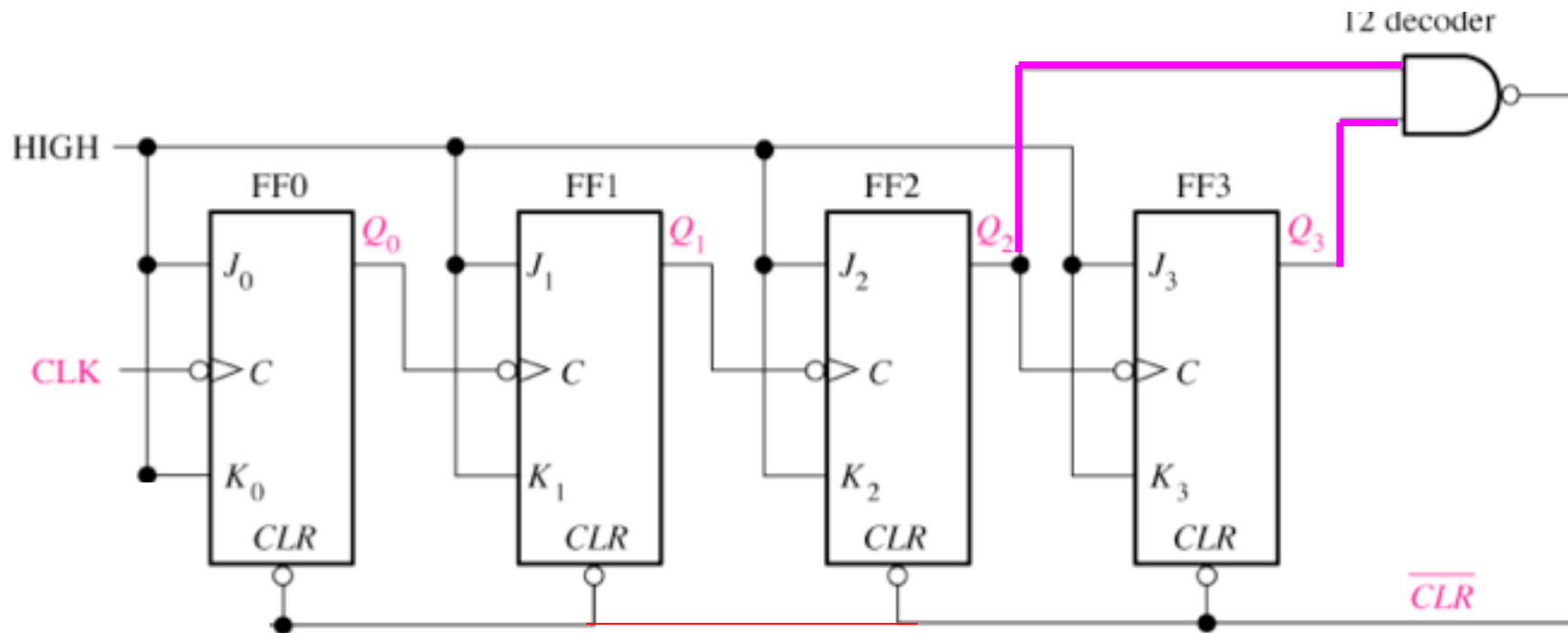
Asynchronously Clocked Modulus-12

Decode twelve (1100_2) with a NAND gate and connect the output to the CLEAR inputs of all flip-flops.

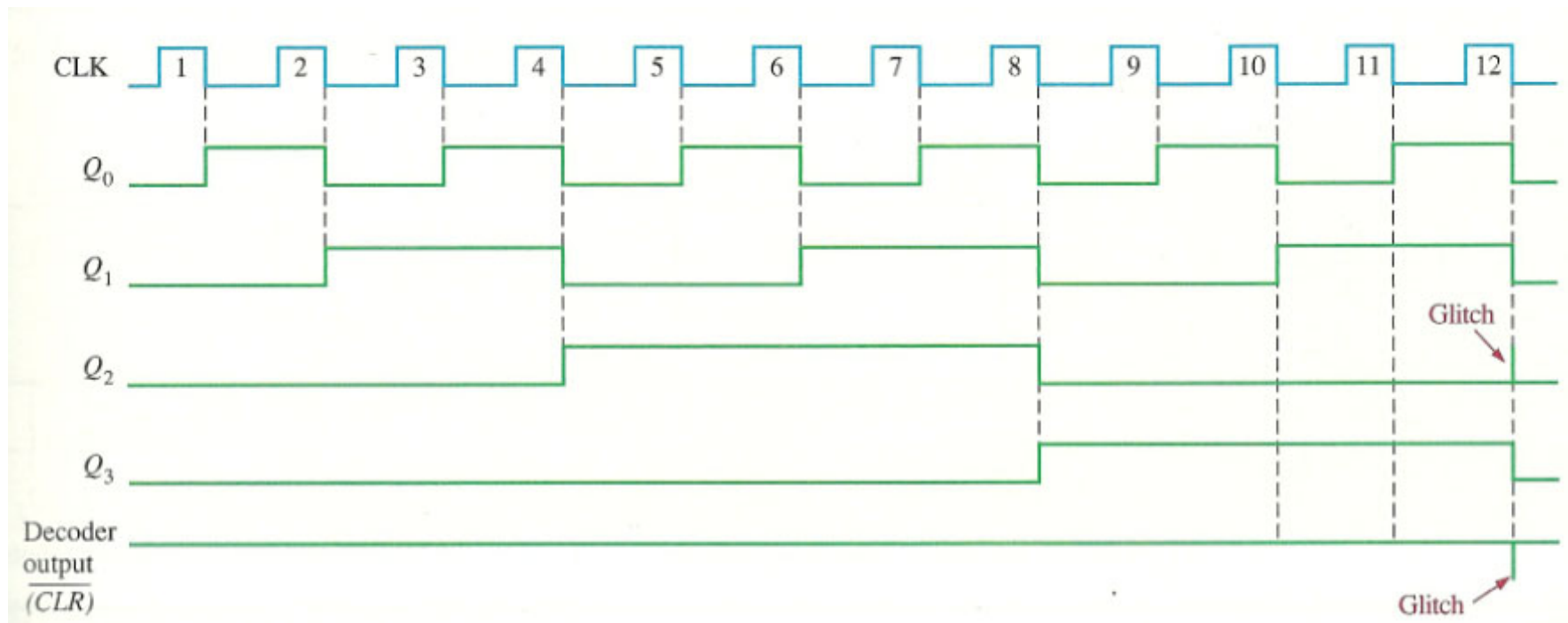
Q_3 Q_2 Q_1 Q_0
1 1 0 0



Decoder Design



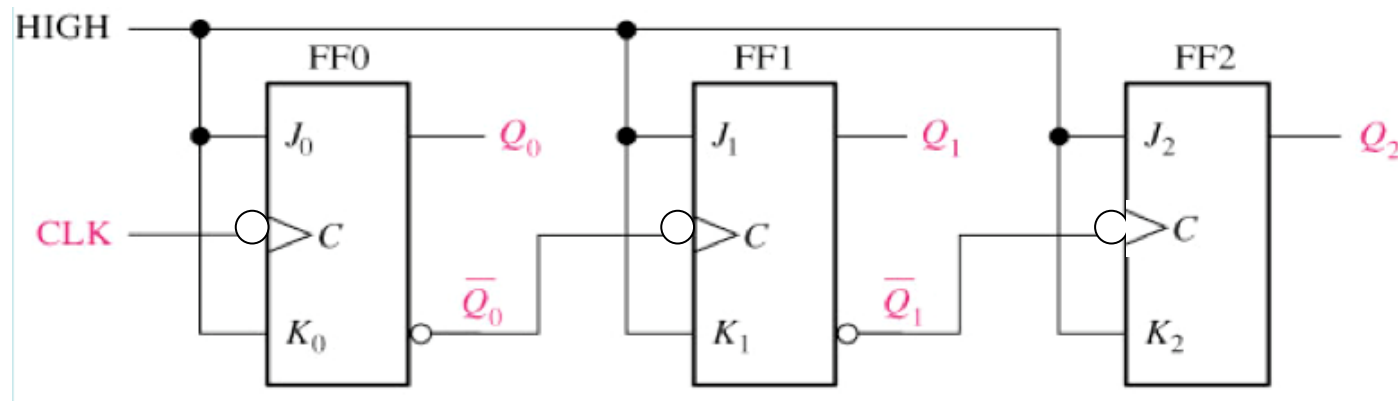
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Exercise 8a.5:



Given a counter designed using J-K flip-flop.



- What is the counting sequence?
- What is the maximum counter state?
- If the counter need to become MOD 5, design the decoder.
- Draw the counter with the decoder in (c).
- Draw the waveform outputs for 8 clock cycles. Assume the initial value of $Q_i = 0$.
- Draw the state diagram the the counter.



Exercise 8a.5b:

Consider a 4-bit count up asynchronous counter is designed using J-K flip-flop with positive edge triggered clock.

- a) If the counter need to become MOD 7, design the decoder.
- b) Draw the counter with the decoder.
- c) Draw the waveform outputs for 8 clock cycles. Assume the initial value of $Q_i = 0$.
- d) Draw the state diagram of the counter.

Advantage and Disadvantage of AC

- Advantage:
 - The design step is simple (easy)
- Disadvantage:
 - Propagated delay accumulate as in ripple binary adder which may cause a missing counting state
 - Especially at high speed (frequency) operation, therefore this kind of counter cannot operates at high frequency.
 - The maximum operating frequency $f_{\max} = \frac{1}{Nt_{pd}}$ where N is the number of flip-flop and t_{pd} is the propagation delay of the flip-flop

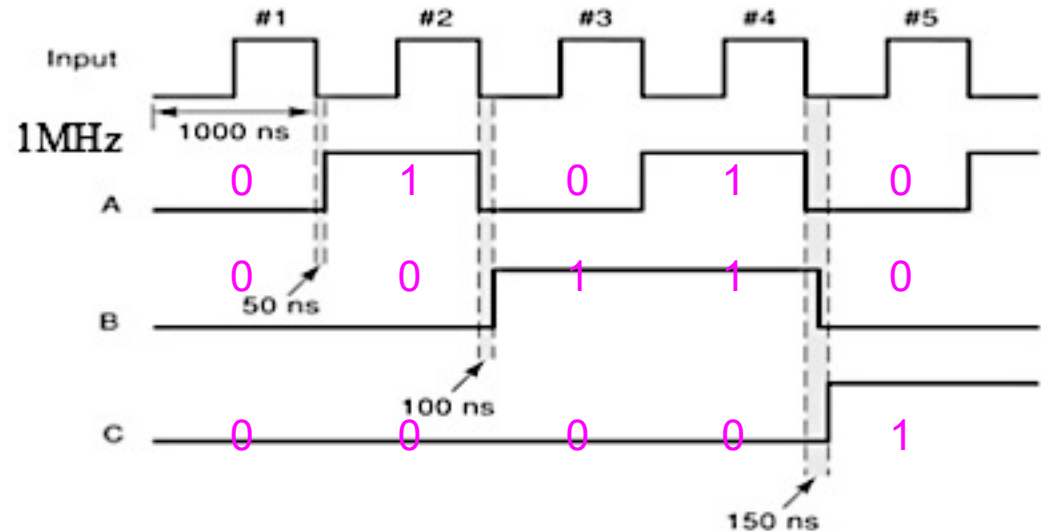
Effect of propagation delay on AC

Assume 3 bits counter using a flip-flop with a propagation delay of 50ns.

Low frequency

Case 1 : operating at low frequency of 1 MHz

For each stage there is a 50 ns delay and it accumulates up to 150ns at the 3rd stage. The accumulated delay is still lower than the period of the signal which is 1000ns, therefore there is no effect to the counting sequence.

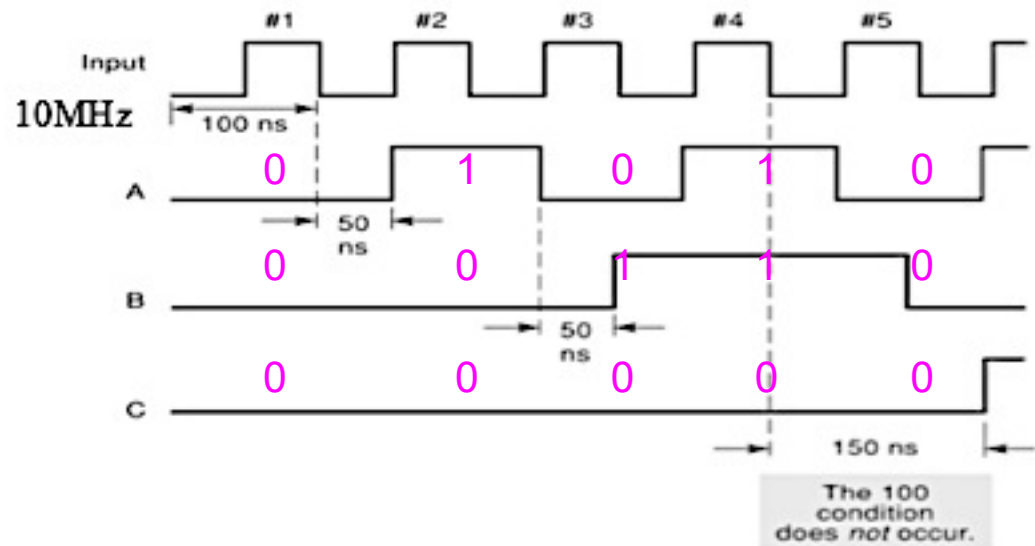


continue...

High frequency

Case 2: operating at a higher frequency of 10 MHz

The clock period is only 100ns. For a 3 stage flip-flop the accumulated delay is the same as before which is 150ns. But, now the accumulated delay is more than the clock period, therefore there is a missing count, 100 doesn't exist!



Exercise 8a.6: If a propagation delay of a mod 32 asynchronous recycle UP counter using JK FF is 40ns, what is the maximum operating frequency?

Solution:

$$\text{Number of flip - flops} = \left\lceil \frac{\log \text{MOD}}{\log 2} \right\rceil$$

$$= \frac{\log 32}{\log 2}$$

$$= \frac{\log 2^5}{\log 2}$$

$$= \frac{5 \log 2}{\log 2}$$

$$N = 5$$

$$f_{\max} = \frac{1}{N t_{pd}}$$

$$= \frac{1}{5(40\text{ns})}$$

$$= \frac{1}{5(40 \times 10^{-9}\text{s})}$$

$$f = 5 \text{ MHz}$$